Mitigating Odor and Greenhouse Gas Emissions in Composting: Effect on Microbial additive

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Introduction

- Livestock manure poses environmental concerns due to emissions, including odorous and greenhouse gases (OECD-FAO Agricultural Outlook 2022-2031).
- Composting offers a sustainable solution by transforming livestock manure into nutrient-rich organic fertilizers.
- Optimizing the composting process involves the introduction of microbial cultures, which enhance decomposition, reduce odorous compounds, and improve compost quality (Sen *et al.*, 2012; Rastogi *et al.*, 2020; Pot *et al.*, 2021).
- Therefore, this research aims to determine the mitigation effect of microbial addition on gas emissions during dairy cattle and swine manure composting.

Objective

This research aims to determine the mitigation effect of microbial additives on ammonia (NH_3), greenhouse gases (CH_4 and N_2O), volatile fatty acids (VFAs), volatile organic compounds (VOCs), and sulfur compounds during dairy cattle and swine manure composting.

Materials and methods

- **Materials**

- Reactor volume : 3.8 L
- Raw materials : Swine manure, Dairy cattle manure and sawdust
- Aeration rate : 0.4 L/min kg-VS (First week)
  0.3 L/min kg-VS (After the first week)
- Microbial strain : *Aquamicrobium lusatiense* NLF
- Gas Analysis : NH_3, CH_4, N_2O, VFAs (Acetic acid, Propionic acid, Isobutyric acid, Butyric acid, Isovaleric acid, Valeric acid), VOCs (Phenol, p-cresol, Indole, and Skatole), Sulfur compounds (MM, H_2S, DMS, and DMDS).

- **Methods**

Swine manure : Cattle manure : Sawdust
5 : 4 : 1

Microbial additive (w.b.)

- Control: 0% microbial
- Treatment-1: +1% microbial (0 days)
- Treatment-2: +0.5% microbial (0 days), +0.5% microbial (14 days)

Gas Analysis

- NH_3, CH_4, CO_2, N_2O
  - 1st week : Every day
  - 2nd week : 2-day interval
  - 3rd week~ : 3-day interval
- VFA, VOC, sulfur compounds and Odor Intensity
  - 7-day interval

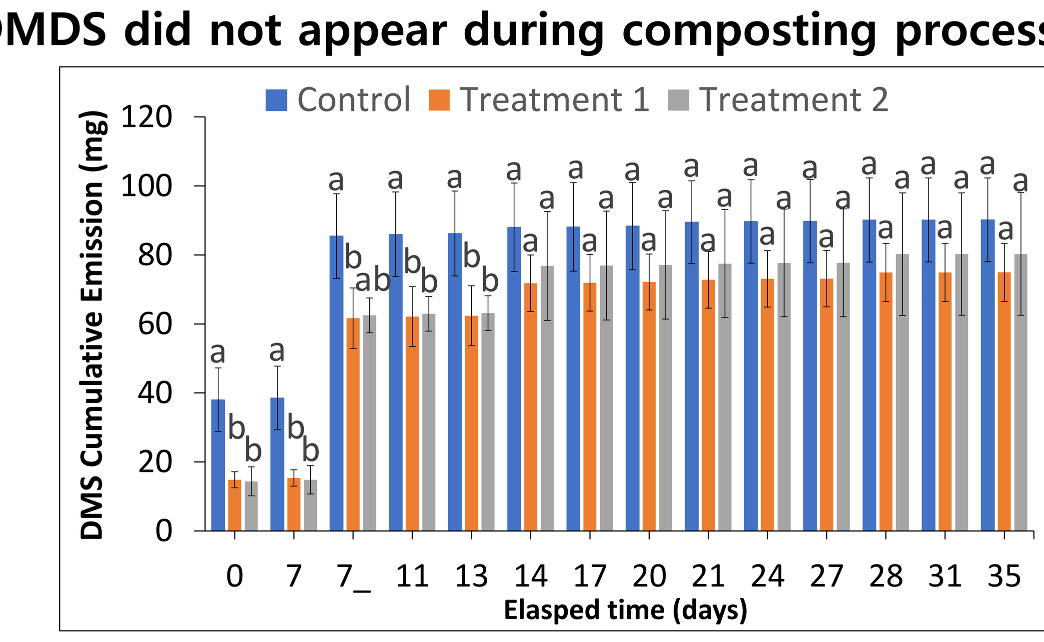
Results and Discussion

- **Greenhouse gas (N_2O and CH_4)**
- The results showed there was no significant difference between the control and the treatment groups ($p>0.05$).
- The cumulative emission of CH_4 were 7.8±3.8g, 8.8±0.46g, and 6.7±2.3g while cumulative emission of N_2O were 105.4±3.4mg, 104.8±19.0mg, and 115.5±4.0mg at control, treatments-1, and treatment-2, respectively.
- **Odor**
  - **NH_3**
- There was no significant difference between the control and the treatment groups ($p>0.05$).
- The cumulative emissions were 5.7±0.0g, 5.7±0.2g, and 5.5±0.2g at control, treatments-1, and treatment-2, respectively.

Results and Discussion

Sulfur compounds (H_2S, MM, DMS, DMDS)

- In the initial 13 days, Treatment-1 exhibited a 27.6% DMS reduction, while Treatment-2 achieved a 26.7% reduction ($p<0.05$) (Figure 1).
  - The study by Jeong et al. (2023) showed that *Aquamicrobium lusatiense* NLF is a sulfur-oxidizing bacteria that can reduce sulfur compounds.
- No significant difference among experimental groups in H_2S ($p>0.05$). Cumulative emissions were 5.5±4.2mg (control), 2.6±1.2mg (treatment-1), and 2.9±2.2mg (treatment-2).
- MM and DMDS did not appear during composting processes.



*Different superscripts on the same day meaning each group is significantly different ($p<0.05$); 7_: Day 7 After mixing.

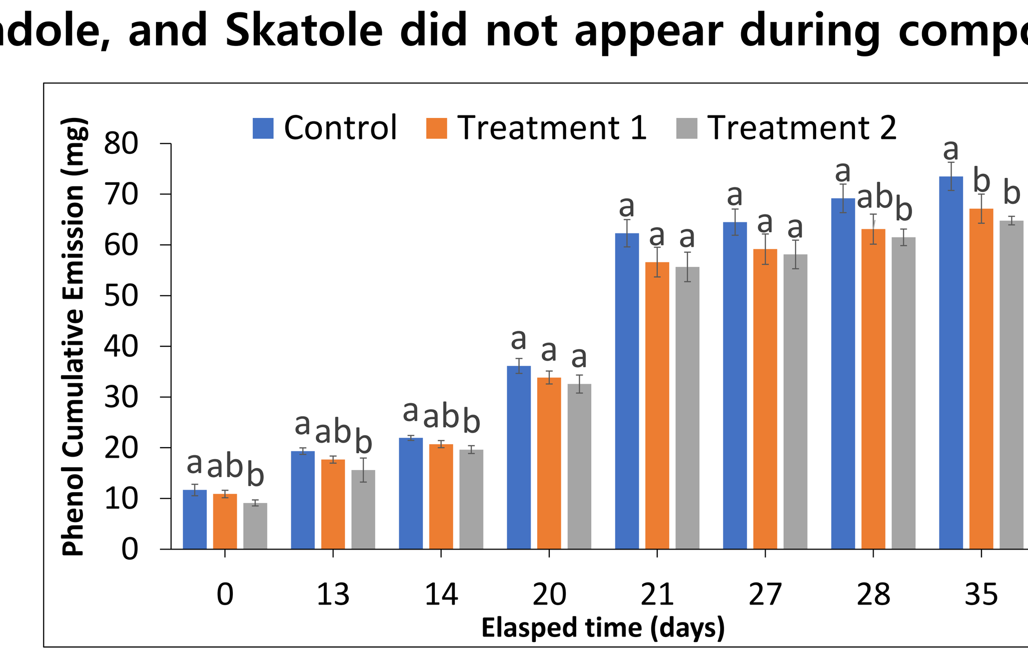
Figure 1. DMS emission during composting

Total VFA

- There were no significant differences ($p>0.05$). Total VFA cumulative emission: control (10.0±0.8mg), treatment-1 (9.6±1.6mg), and treatment-2 (10.8±0.7mg).

VOC (Phenol, p-cresol, Indole, and Skatole)

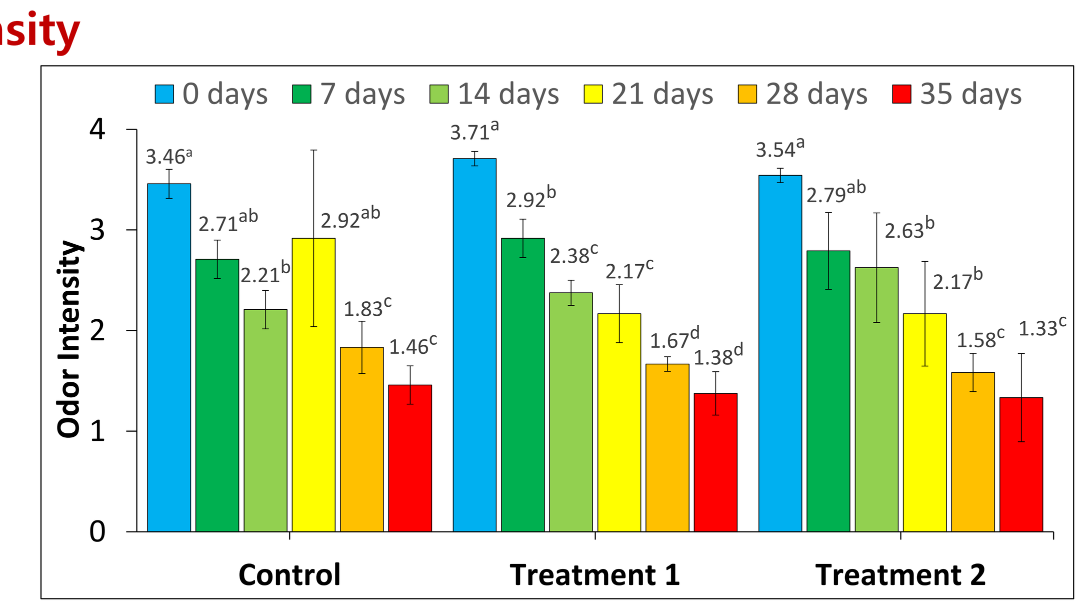
- Treatment-2 showed a significant phenol reduction of 11.7% and Treatment-1 8.6% ($p>0.05\%$) after 35 days.
  - Fristche *et al.*, 1999 reported the capability of *Aquamicrobium lusatiense* to degrade phenol when cultured in the medium.
- P-Cresol, Indole, and Skatole did not appear during composting processes.



*Different superscripts on the same day meaning each group is significantly different ($p<0.05$)

Figure 2. Phenol emission during composting

Odor Intensity



*Different superscripts on the same treatments meaning each group is significantly different ($p<0.05$).

Figure 3. Odor Intensity results

- Over 35 days, control reduced from 3.46 to 1.46, Treatment-1 from 3.71 to 1.38, and treatment-2 reduced from 3.54 to 1.33.

Conclusions

- Sulfur oxidizing activity by *Aquamicrobium lusatiense* NLF was observed at the early stages of composting.
- The significant mitigation effect on phenol is observed due to the capability of *Aquamicrobium lusatiense* NLF to use phenolic compounds as carbon sources.
- Future research is needed on refining these treatments and assessing their extended impacts on compost quality and microbial communities.

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